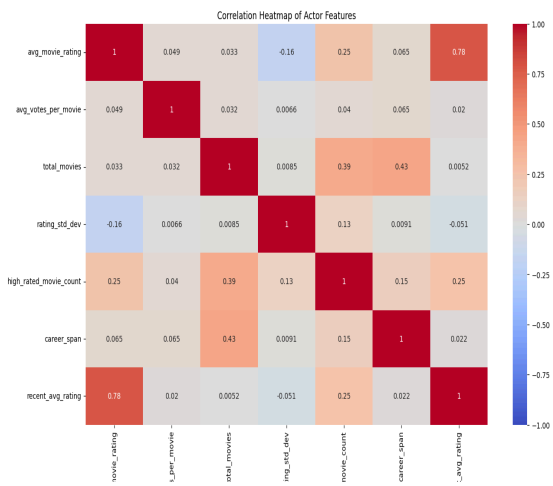
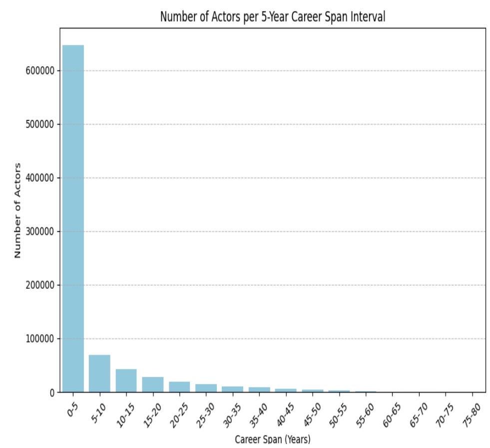
This investigation quantified the impact of actors on movie ratings using the IMDb non-commercial database. The core of the analysis is the actor scores table, which captures actors' performance and experience through features like average movie rating, average votes per movie, total movies, movie rating standard deviation, high-rated movie count, career span, and recent average rating (last 5 years). A correlation heatmap of these features helped identify relationships. In Figure 1, it is shown that there is a high correlation between the average movie rating and recent average movie ratings, suggesting future models should use only one of the features.





Data preprocessing addressed quality issues in the dataset. Actors with fewer than three movies (650,000 with one movie, 110,000 with two) were excluded to reduce noise, as their limited features skewed the data, confirmed by right-skewed distributions in plots of actors per 5-year career span and total movies in Figure 2. Of the remaining 222,531 actors, 17,199 actors had missing rating standard deviation values due to having only one rated movie; these were imputed using the average rating standard deviation for actors with a 0–5 year career span. Additionally, 142,180 actors lacked a recent average rating (no movies since 2020), and were imputed with the average across all actors. Actors with career spans over 80 years (142, mostly early 20th-century actors with missing death dates) were removed to eliminate data errors.

The cleaned dataset was used to train a Ridge regression model on movies from 2020 to 2025, mitigating the risk of inflated actor scores for older movies (e.g., from 10 years ago) due to using all historical data. Analysis of the IMDB title_principals table confirmed that higher ordering (1–2) corresponds to leading actors, while lower ordering (3–7) indicates supporting cast, validating the aggregation of scores for the top 7 actors. The average actor score of these actors formed each movie's average actor score feature, representing cast quality. The Ridge regression model achieved an MSE of 0.6638, RMSE of 0.8148, and R^2 of 0.7283, explaining 72.83% of the variance in movie ratings. The model's weights were used to compute each actor's final actor score as a weighted sum of standardized features, scaled and shifted to ensure non-negative values, facilitating interpretable predictions of movie ratings. In the future, the actor scores table will be created using a time-aware approach, where for each movie, actor features will contain information about the actor until the movie is released. Aggregate actor scores per movie will be used to train the model on a collection of movie-related features, such as genre, directors, budget, etc.